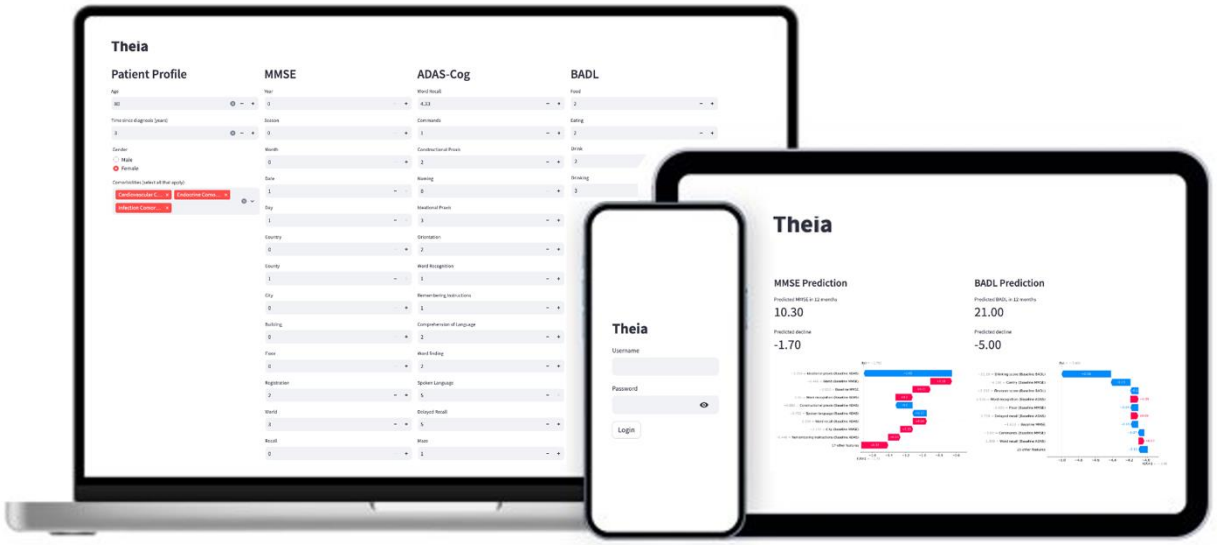


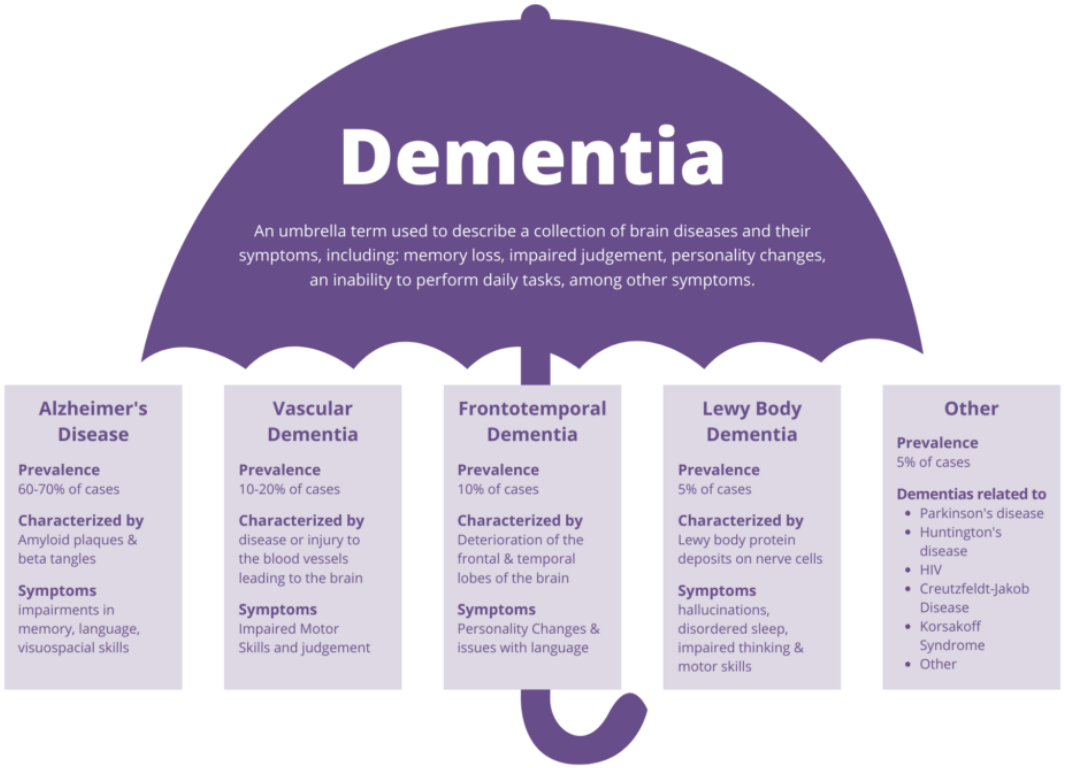
Predicting dementia progression using pen and paper clinical assessments

Antigone Fogel
Department of Brain Sciences & UK DRI Care Research and Technology Centre

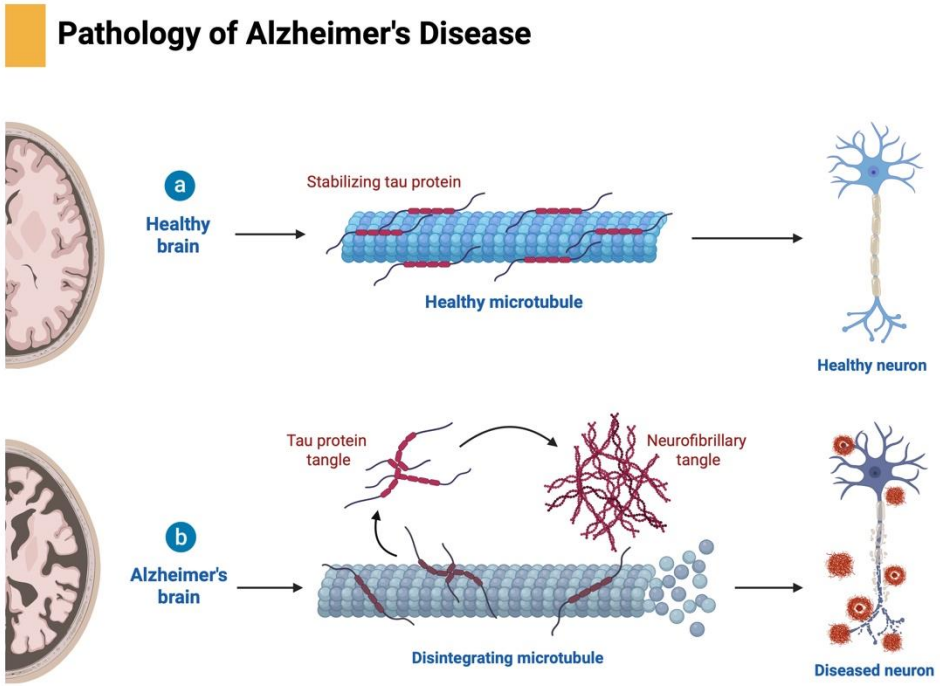
UK Dementia Research Institute

IMPERIAL

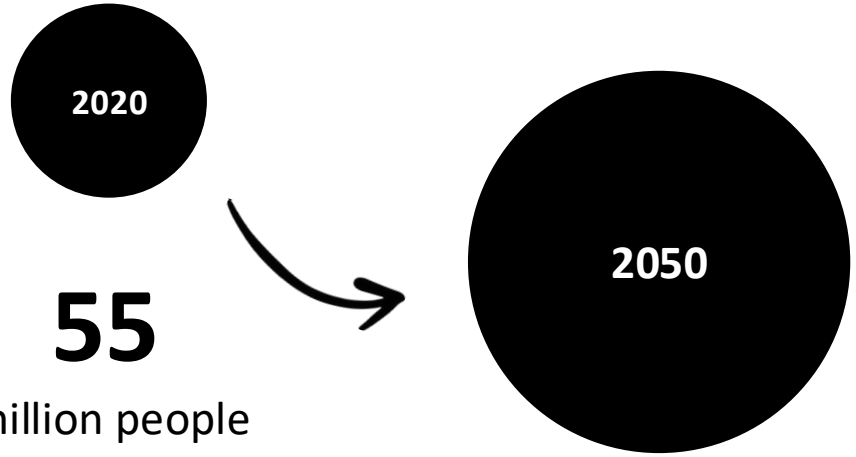
Dementia



Source: https://coahamilton.ca/2022/02/what-does-my-diagnosis-mean/



Dementia



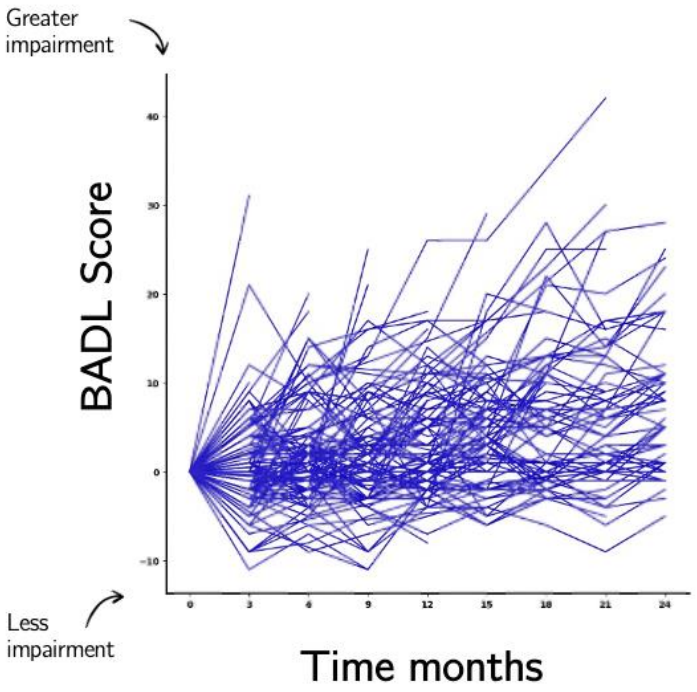
55
million people

139
million people

Key question: “What happens next?”

The problem

Current prognosis methods rely on **clinician experience** and **heuristics**



Rates of cognitive and functional decline **vary considerably** across people living with dementia

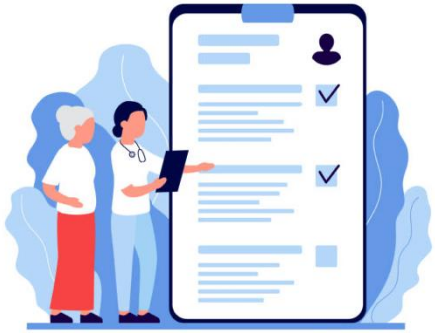
The problem:

There are **no tools** to support development of **personalised dementia prognoses**

The problem:

There are **no tools** to support development of **personalised dementia prognoses**

Why do personalised prognoses matter?



Tailored care

Emotional support

Background:

Machine learning in medicine

Why use machine learning in medicine?

ML allow us to:

1. Learn from a large number of past patients' medical histories
2. Offer personalised predictions and care suggestions
3. Provide mechanistic insights

Machine learning in medicine: a balancing act

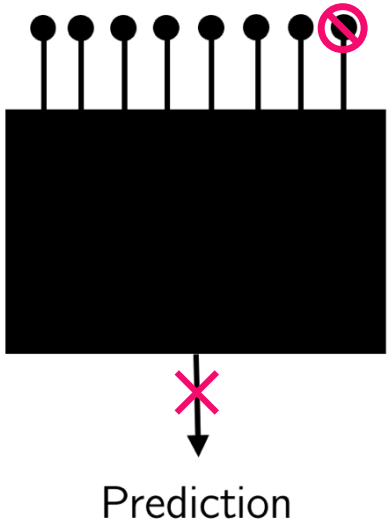
machine learning

medicine

Considerations when using machine learning for medical applications

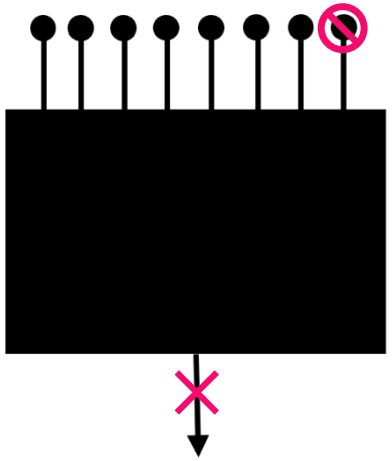
Considerations when using machine learning for medical applications

1 Accessible inputs

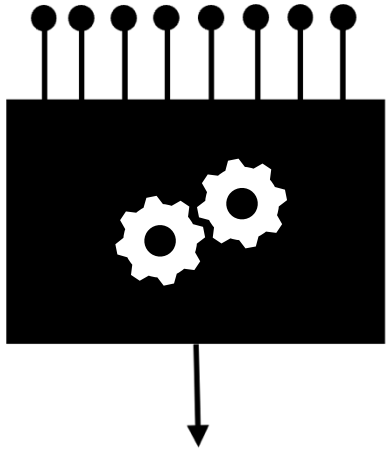


Considerations when using machine learning for medical applications

1 Accessible inputs

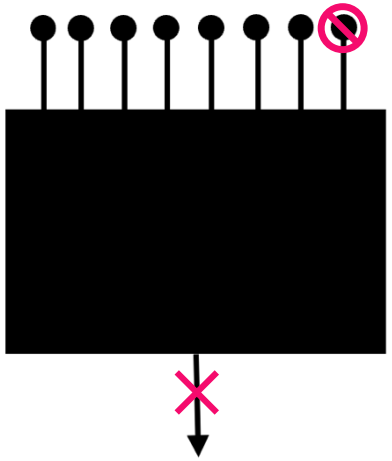


2 Explainable

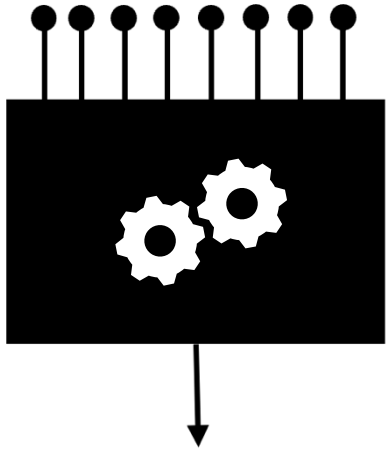


Considerations when using machine learning for medical applications

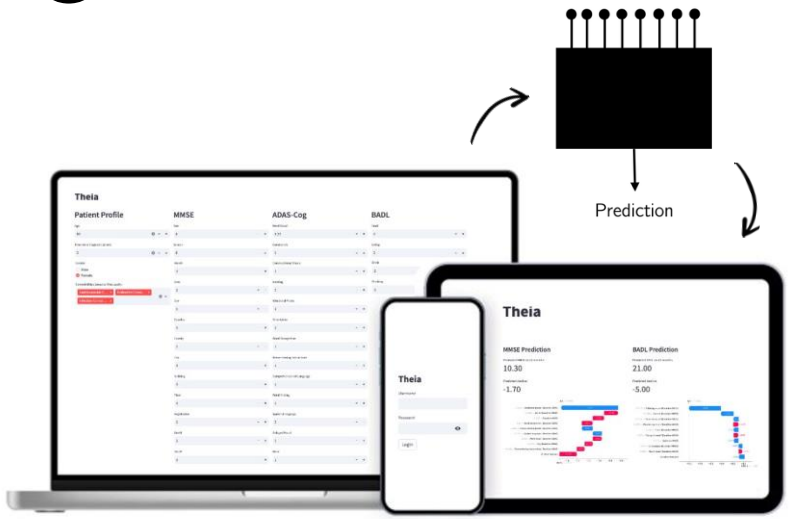
1 Accessible inputs



2 Explainable



3 Useful and easy to use



Methodology:

How do we model dementia progression?

Methodology

Existing predictive models of dementia progression

Received: 26 April 2023 | Revised: 6 September 2023 | Accepted: 7 November 2023
DOI: 10.1002/alz.13565
Alzheimer's & Dementia
THE JOURNAL OF THE ALZHEIMER'S ASSOCIATION

RESEARCH ARTICLE

Predicting clinical progression trajectories of early Alzheimer's disease patients

Viswanath Devanarayan
Pallavi Sachdev[1] | Dan
Lynn Kramer[1] | Shobh
Neuroimaging Initiative (

Received: 6 September 2019 | Revised: 21 October 2019 | Accepted: 25 November 2019
DOI: 10.1002/alz.12032
Alzheimer's & Dementia
THE JOURNAL OF THE ALZHEIMER'S ASSOCIATION

FEATURED ARTICLE

Predicting sporadic Alzheimer's disease progression via inherited Alzheimer's disease-informed machine-learning

Nicolai Franzmeier[1] | Nikolaos Koutsouleris[2] | Ta
Alison Goate[5,6] | Celeste M. Karch[4,7,8] | Anne M. Fa
Marco Duering[1] | Martin Dichgans[1,10,11] | Johannes
Brian A. Gordon[4,13,14] | Yen Ying Lim[15] | Colin L. Ma
Nick C. Fox[16] | Antoinette O'Connor[16] | Jasmeer Ch
Adrian Danek[12] | Jason Hassenstab[4,9,14] | Peter R. S
Randall J. Bateman[4,9] | the Alzheimer's disease neuro
Dominantly Inherited Alzheimer Network (DIAN)[22] |

JAMA Neurology | Original Investigation

Prediction of Longitudinal Cognitive Decline in Preclinical Alzheimer Disease Using Plasma Biomarkers

Niklas Mattsson-Carlgren, MD, PhD; Gemma Salvadó, PhD; Nicholas J. Ashton, PhD; Pontus Tideman, MSc; Erik Stomrud, MD, PhD;
Henrik Zetterberg, MD, PhD; Rik Ossenkoppele, PhD; Tobey J. Betthauser, PhD; Karly Alex Cody, BS; Erin M. Jonaitis, MS, PhD; Rebecca Langhough, PhD;
Sebastian Palmqvist, MD, PhD

nature communications
Article
https://doi.org/10.1038/s41467-022-35712-5

Forecasting individual progression trajectories in Alzheimer's disease

holand[1], Colin Birkenbihl[2,3],
Stéphane Epelbaum[7],
[2,3] & Stanley Durrleman[1]

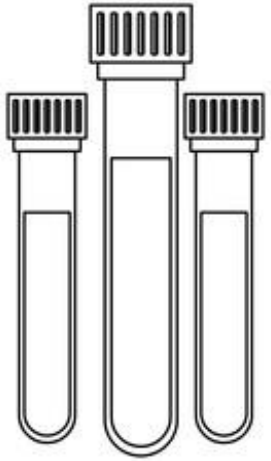
Diagnostic Assessment & Prognosis

Characterizing heterogeneity in the progression of Alzheimer's disease using longitudinal clinical and neuroimaging biomarkers

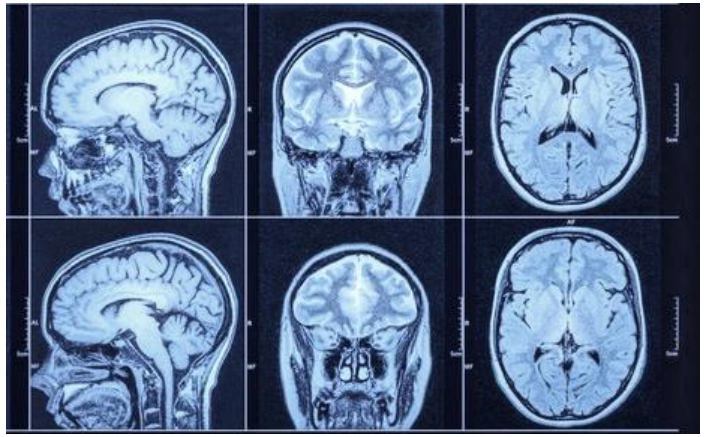
Devendra Goyal[a], Donna Tjandra[a], Raymond Q. Migrino[c], Bruno Giordani[b], Zeeshan Syed[d], Jenna Wiens[a,*], for the Alzheimer's Disease Neuroimaging Initiative[1]

[a]Computer Science and Engineering, University of Michigan, Ann Arbor, MI, USA
[b]Neuropsychology Section, Department of Psychiatry, University of Michigan, Ann Arbor, MI, USA
[c]Phoenix Veterans Affairs Health Care System, Phoenix, AZ, USA
[d]School of Medicine, Stanford University, Stanford, CA, USA

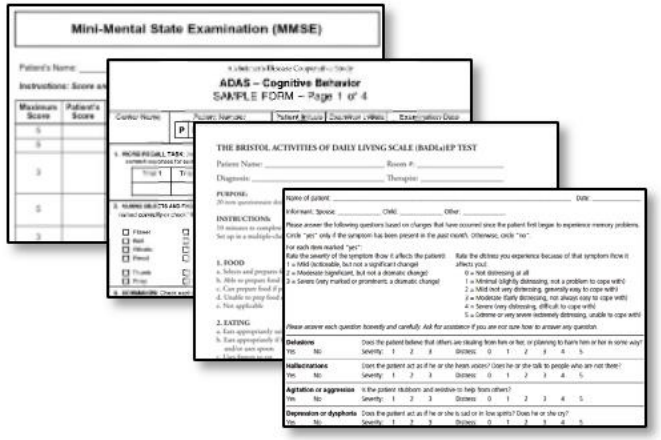
Existing biomarkers of dementia progression



Fluid Biomarkers



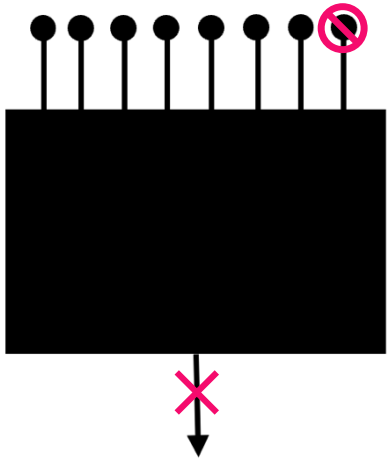
Neuroimaging Biomarkers



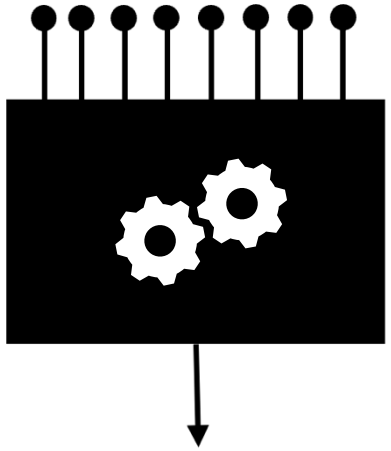
Clinical Assessments

Considerations when using machine learning for medical applications

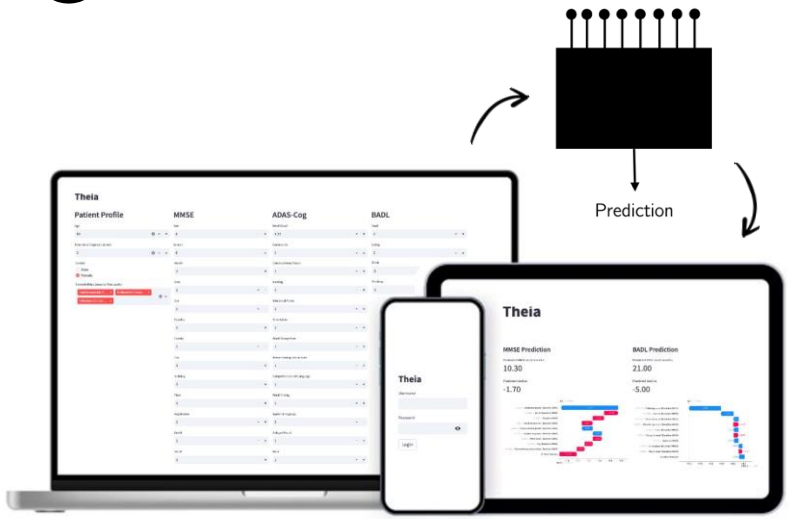
1 Accessible inputs



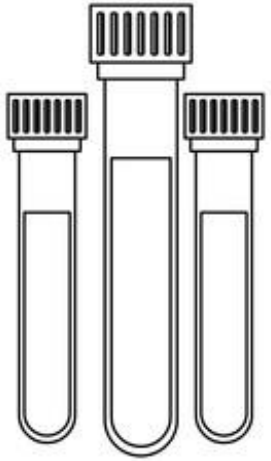
2 Explainable



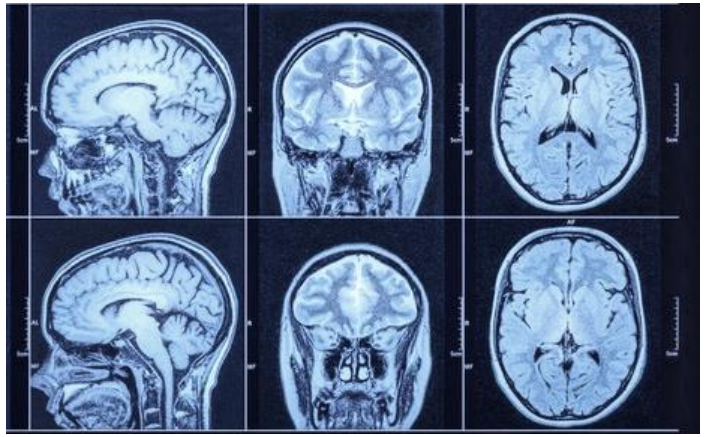
3 Easy to use



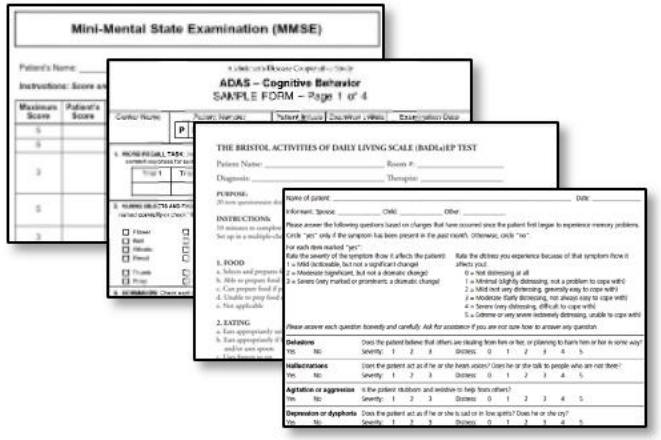
Existing biomarkers of dementia progression



Fluid Biomarkers



Neuroimaging Biomarkers

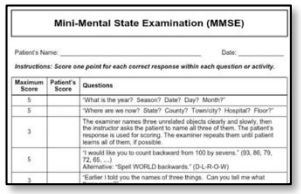


Clinical Assessments

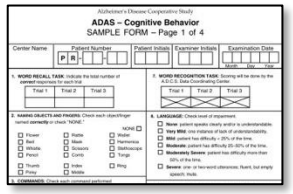
Model inputs

Minder
By UK Dementia Research Institute

MMSE
Mini Mental State Exam

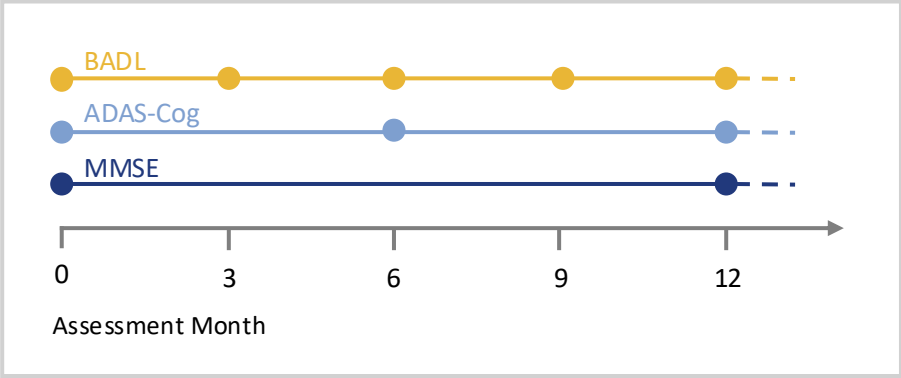

Mini-Mental State Examination (MMSE)

ADAS-Cog
Alzheimer's Disease Assessment Scale


ADAS – Cognitive Behavior
SAMPLE FORM – Page 1 of 4

BADL
Bristol Activities of Daily Living

THE BRISTOL ACTIVITIES OF DAILY LIVING SCALE (BADLeEP TEST



Clinical Assessment Timeline

Age & Gender

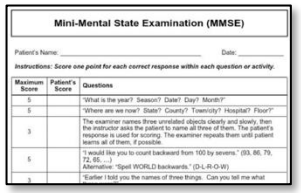
Comorbidities

Prediction targets

Cognitive function

MMSE

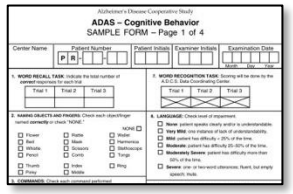
Mini Mental State Exam



Target: 12-month MMSE

ADAS-Cog

Alzheimer's Disease Assessment Scale

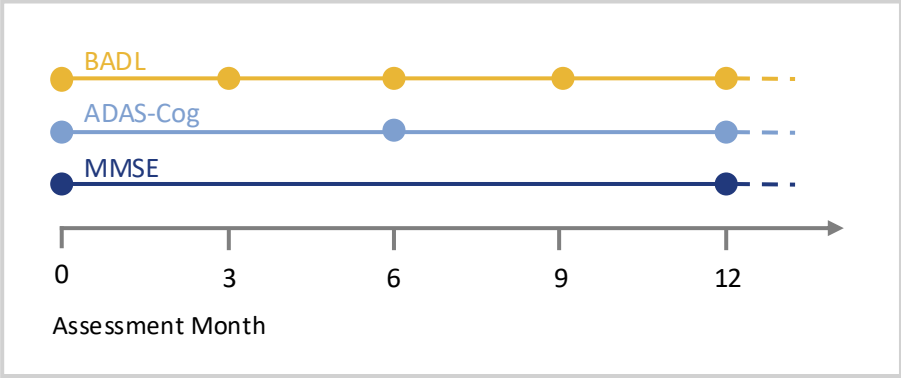


Daily independence

BADL

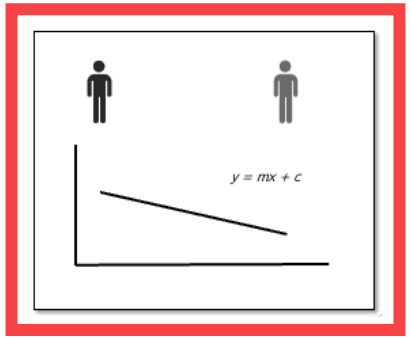
Bristol Activities of Daily Living

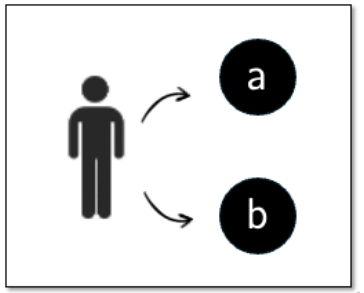
Target: 12-month BADL



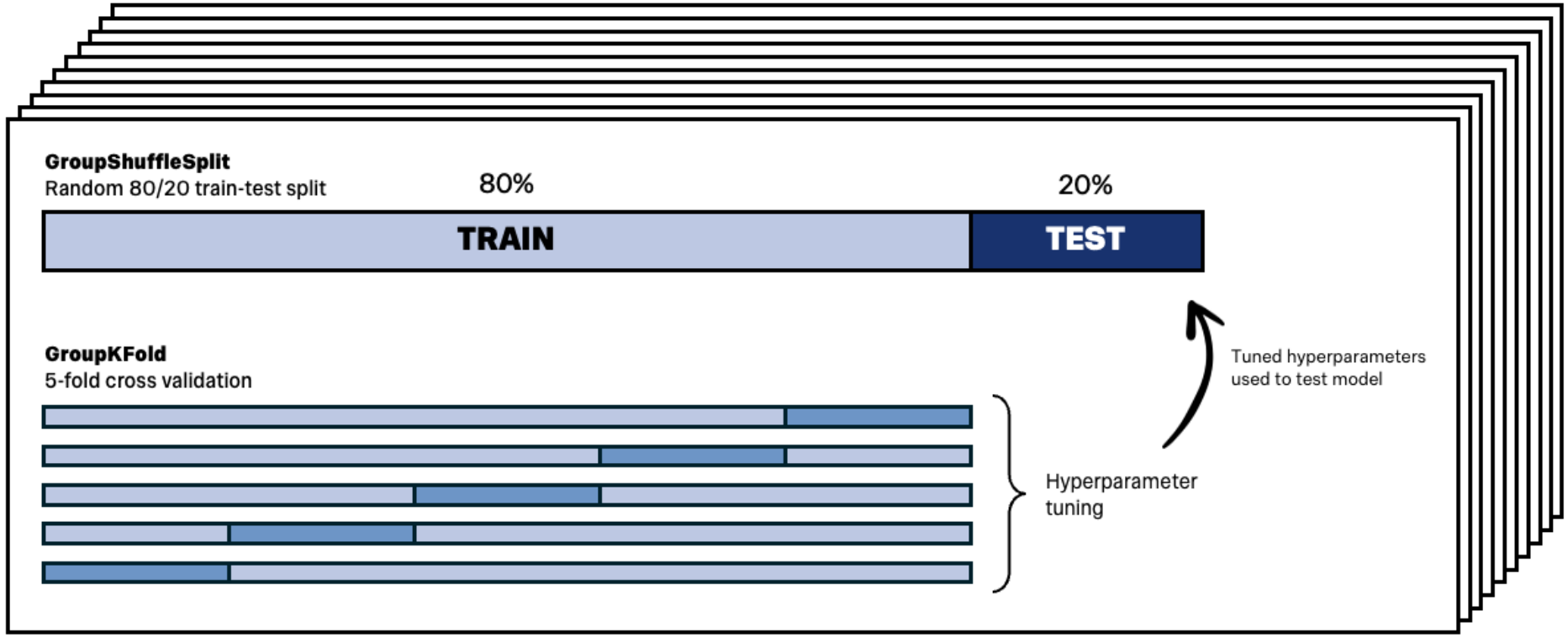
Clinical Assessment Timeline

How do we model progression?





Predictive modelling approach



Predictive model performance

MMSE: 0-30
BADL: 0-60

MMSE Model	MSE (95% CI)	MAE (95% CI)	R^2 (95% CI)
MMSE + ADAS-Cog	5.64 (4.40 – 6.88)	1.85 (1.65 – 2.04)	0.74 (0.67 – 0.81)
MMSE + ADAS-Cog + Comorbidities	**5.60 (4.42 – 6.78)**	**1.84 (1.64 – 2.04)**	**0.74 (0.67 – 0.80)**
MMSE + ADAS-Cog + BADL	7.84 (6.15 – 9.52)	2.28 (1.98 – 2.58)	0.69 (0.57 – 0.80)
MMSE + ADAS-Cog + BADL + Comorbidities	7.83 (6.21 – 9.44)	2.28 (1.99 – 2.56)	0.69 (0.58 – 0.80)
Validation Cohort (ADNI)	8.01	2.19	0.69
BADL Model	MSE (95% CI)	MAE (95% CI)	R^2 (95% CI)
BADL	25.45 (18.47 – 32.42)	3.99 (3.54 – 4.45)	0.75 (0.69 – 0.80)
BADL + MMSE	24.38 (19.48 – 29.29)	3.94 (3.55 – 4.33)	0.75 (0.71 – 0.80)
BADL + ADAS-Cog	**22.93 (17.87 – 28.00)**	**3.88 (3.46 – 4.30)**	**0.77 (0.72 – 0.82)**
BADL + MMSE + ADAS-Cog	23.96 (18.69 – 29.23)	3.93 (3.52 – 4.35)	0.76 (0.71 – 0.81)
BADL + MMSE + ADAS-Cog + Comorbidities	26.64 (20.81 – 32.47)	4.11 (3.64 – 4.58)	0.74 (0.69 – 0.78)

How do models perform with and without imaging data?

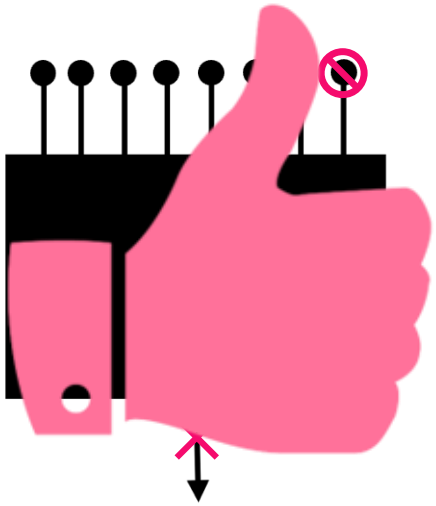
Table S14: Model performance for models of 12-month cognitive decline, with and without neuroimaging features in the ADNI cohort.

Neuroimaging	Model	MSE (95% CI)	MAE (95% CI)	R^2 (95% CI)
Included	$L1$ Regularised Linear Regression	7.09 (6.04 – 8.14)	2.02 (1.88 – 2.16)	0.71 (0.66 – 0.76)
	$L2$ Regularised Linear Regression	7.66 (6.42 – 8.91)	2.13 (1.99 – 2.27)	0.69 (0.63 – 0.75)
	ElasticNet	7.16 (6.08 – 8.24)	2.02 (1.90 – 2.14)	0.71 (0.66 – 0.76)
Excluded	$L1$ Regularised Linear Regression	7.16 (6.08 – 8.24)	2.04 (1.90 – 2.18)	0.71 (0.66 – 0.76)
	$L2$ Regularised Linear Regression	7.47 (6.37 – 8.57)	2.09 (1.96 – 2.23)	0.70 (0.64 – 0.75)
	ElasticNet	7.18 (6.13 – 8.24)	2.04 (1.91 – 2.17)	0.71 (0.66 – 0.76)

MSE: Mean-squared-error, MAE: Mean-absolute-error.

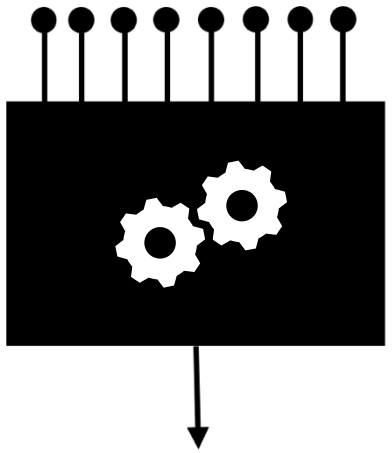
Considerations when using machine learning for medical applications

1 Accessible inputs



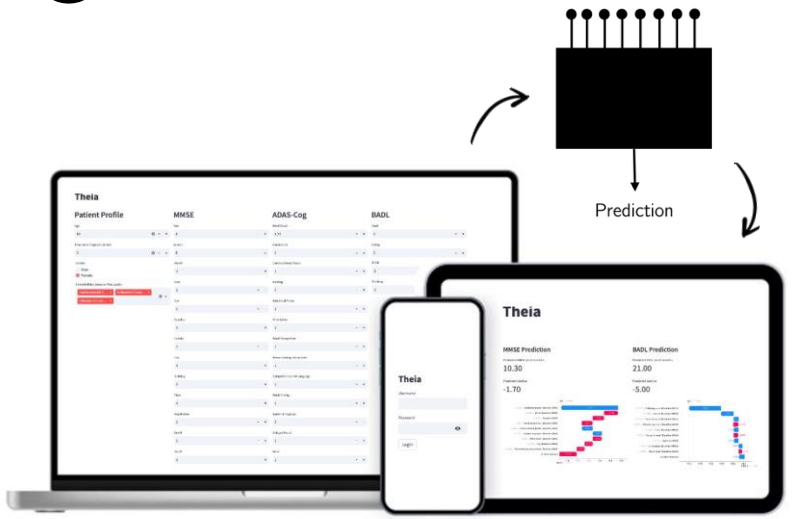
Prediction

2 Explainable



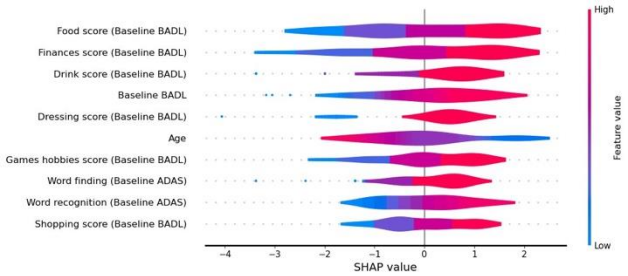
Prediction

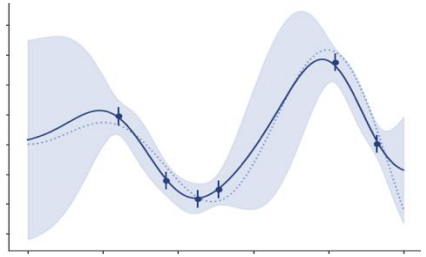
3 Easy to use

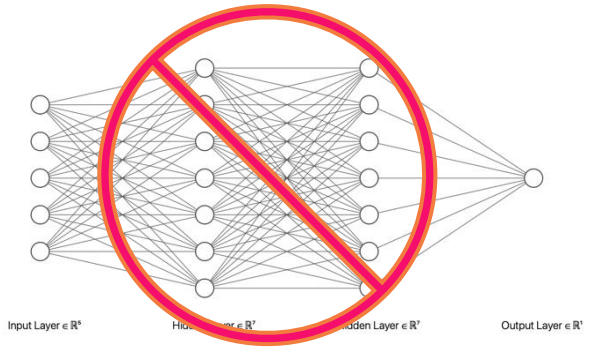


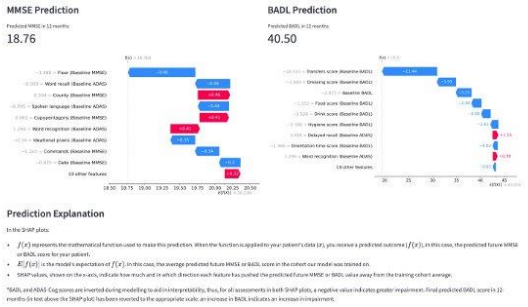
Methodology

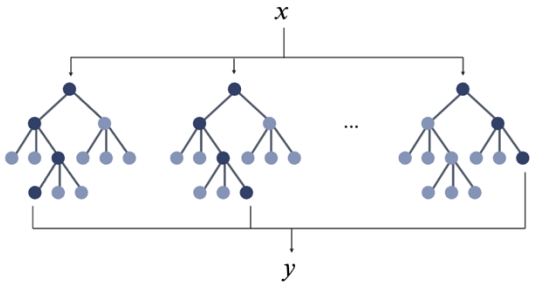
What models should we use?

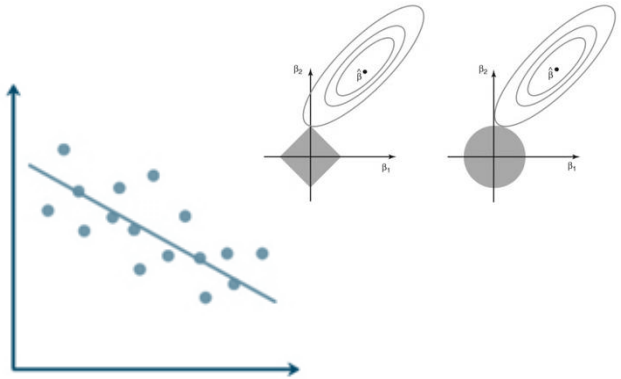




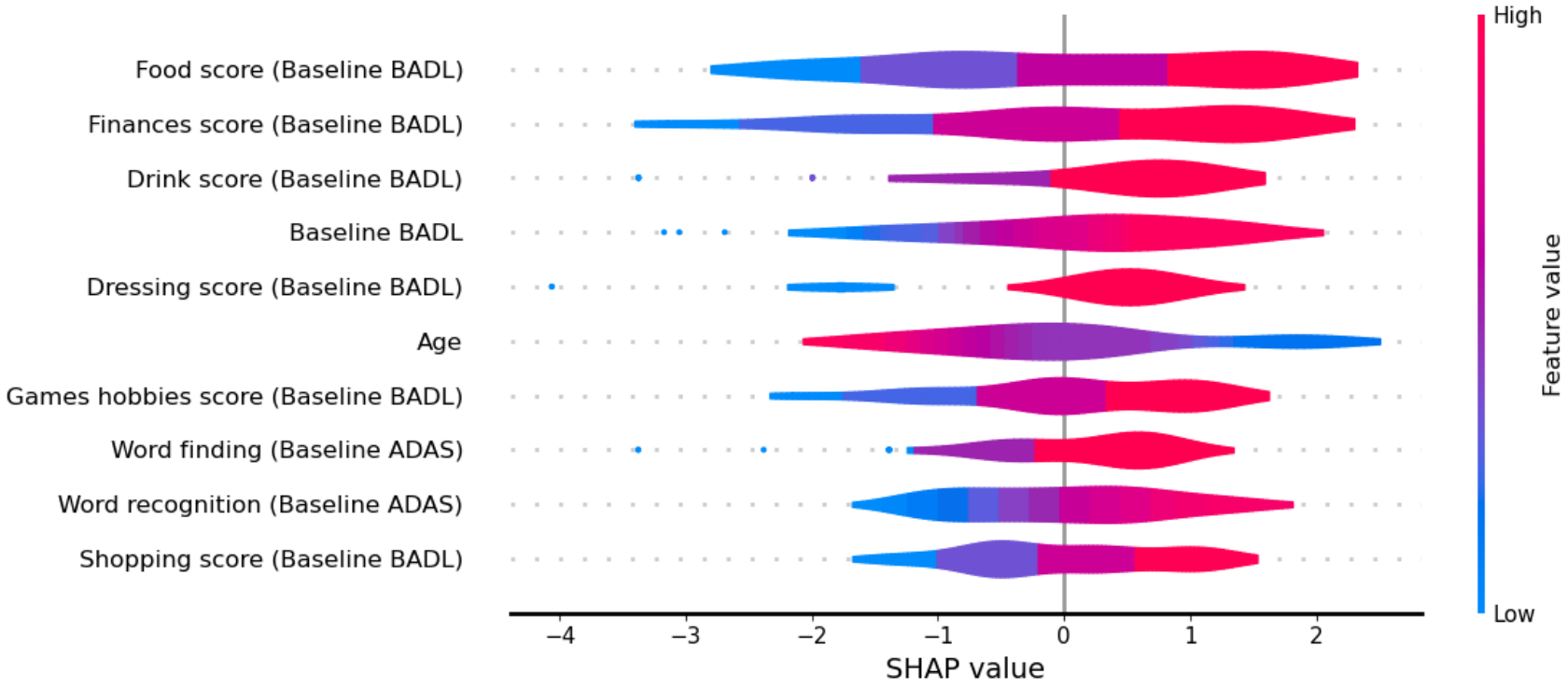






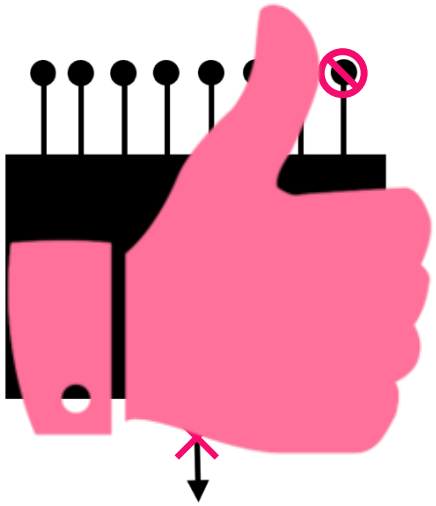


Exploring model decision making: BADL



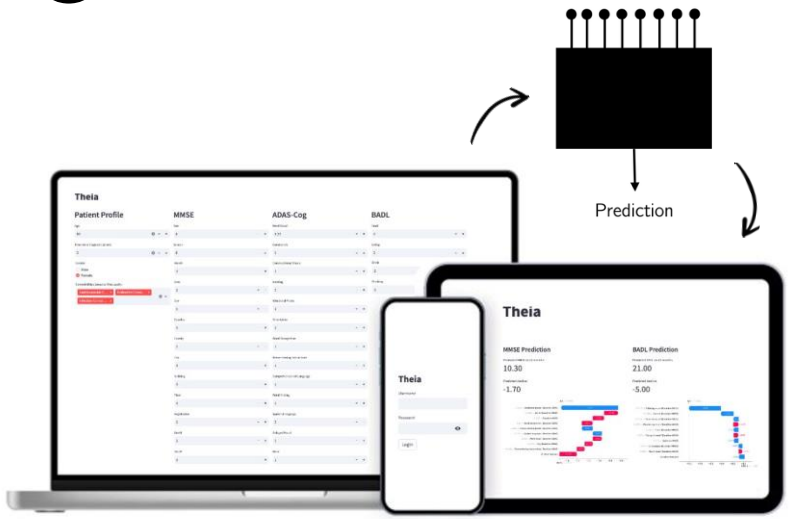
Considerations when using machine learning for medical applications

1 Accessible inputs



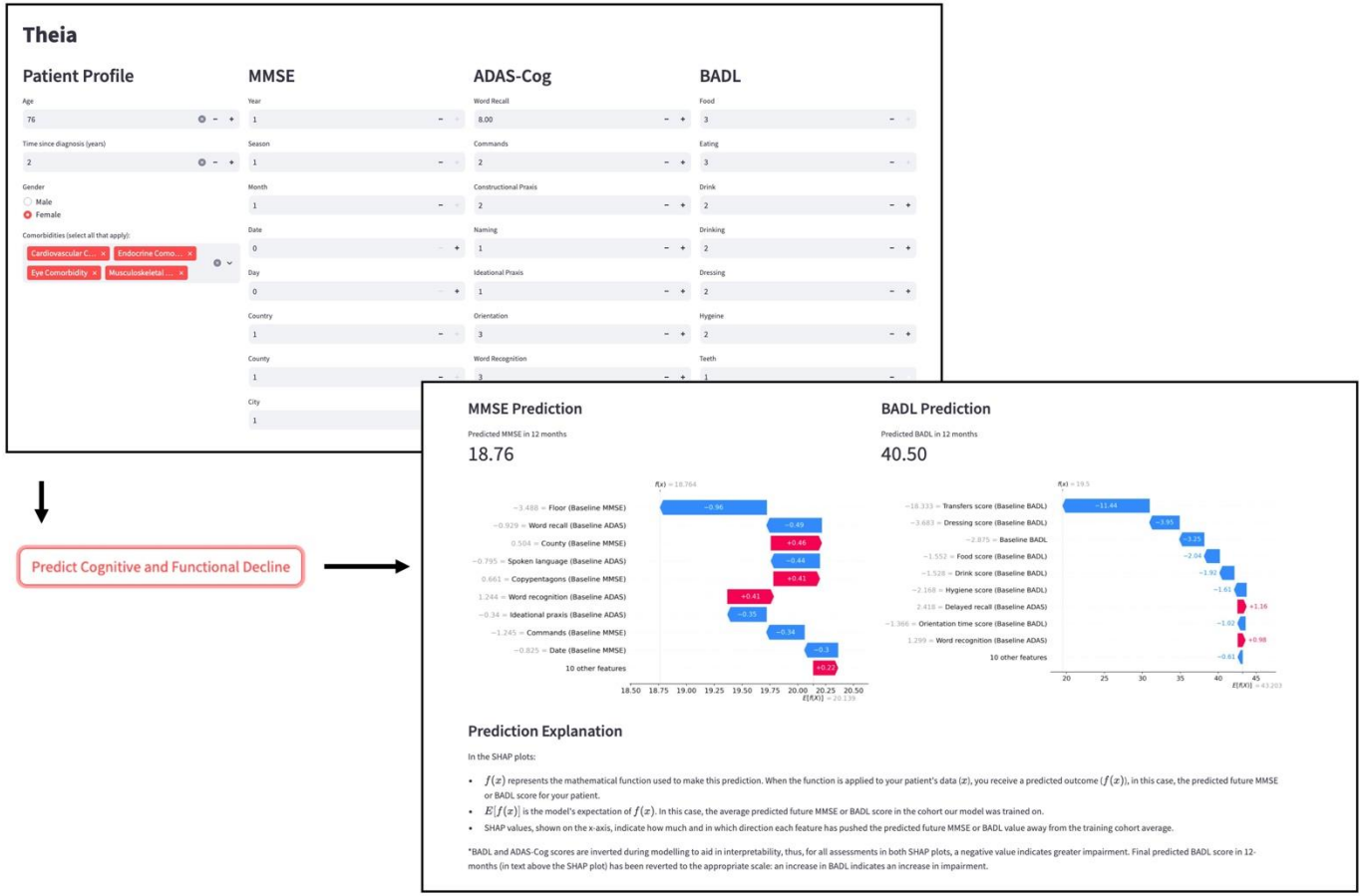
2 Explainable

3 Easy to use



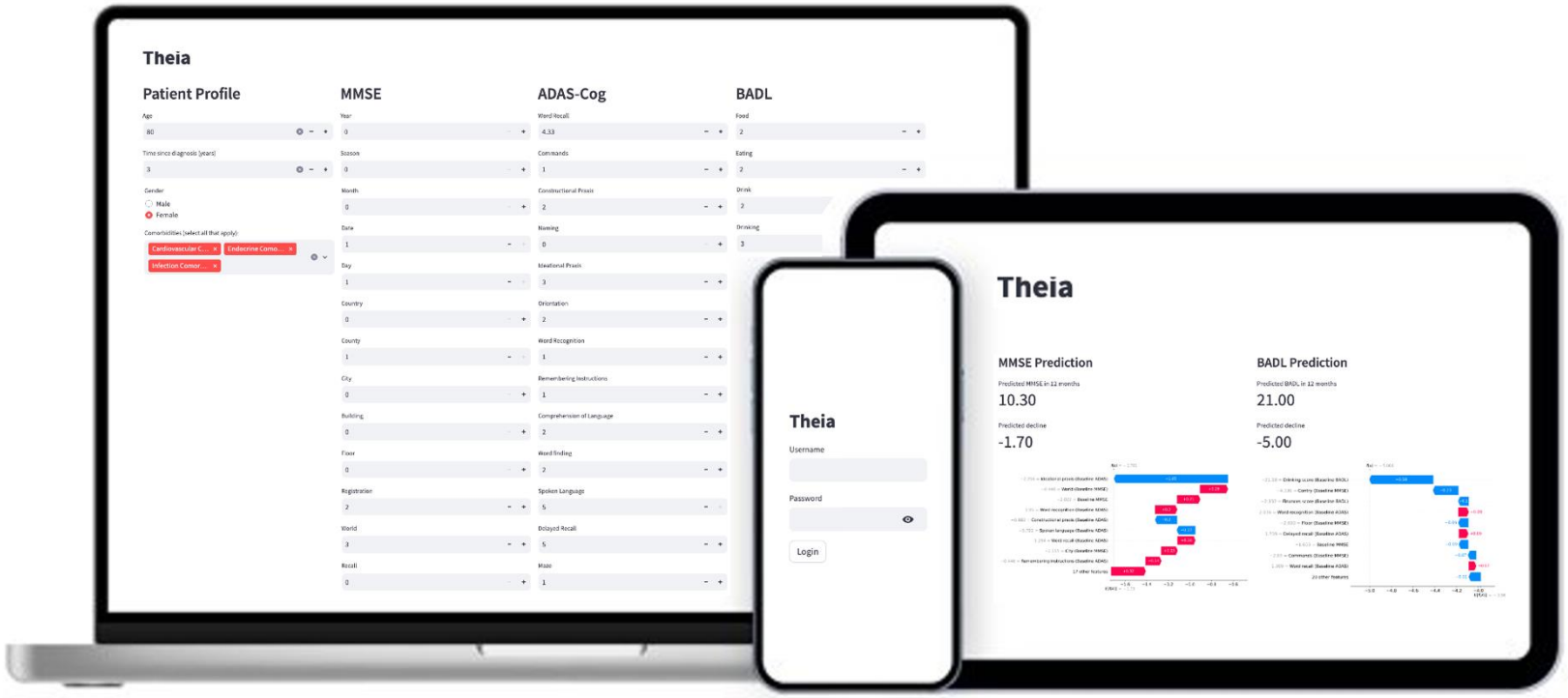
Results

Digital consultation tool & personalised insights



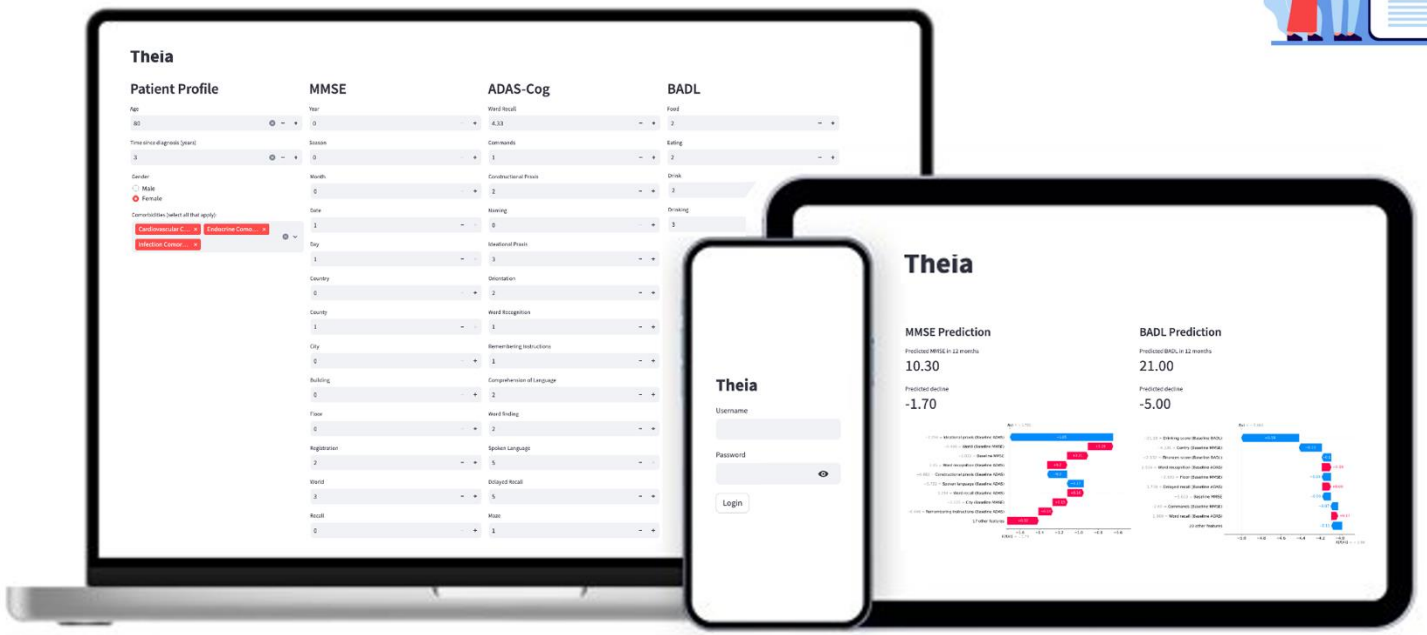
Results

Digital consultation tool & personalised insights



Summary & Implications

Tailored care
Emotional support



- **User friendly** digital consultation tool
- **Readily available** and accessible model inputs
- Deploys an **explainable** model

If you would like to learn more…

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Predicting rates of cognitive and functional decline in Alzheimer's disease and mild cognitive impairment

Antigone Fogel, Chloe Walsh, Nan Fletcher-Lloyd, Paresh Malhotra, Mina Ryten, Ramin Nilforooshan & Payam Barnaghi

Communications Medicine , Article number: (2026) | Cite this article

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Thank you!